20/01/25 M/S

USN | | | | | | | | | | [23CSE110]

FIRST SEMESTER B.Tech EXAMINATION, JANUARY-2025

PROBLEM SOLVING THROUGH PROGRAMMING IN C

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		Questions	Blooms Level	CO's	Marks
		PART-A	**5 x2=10**		
1.	**a.**	What is the difference between RAM and ROM?	L1	CO1	**02**
	b.	What is the purpose of a compiler?	L1	CO1	**02**
	c.	What are tokens in C?	L2	CO2	**02**
	d.	Explain the difference between System Software and Application Software	L2	CO2	**02**
	e.	What is an array in C? Explain how you can declare and initialize a one-dimensional array	L2	CO2	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Describe the basic structure of a computer architecture with respect to its main components	L2	CO2	**05**
		OR			
	b.	Write the algorithm and pseudo code for performing a linear search on an array.	L2	CO2	**05**
3.	**a.**	Describe the if statement and provide an example program.	L2	CO3	**05**
		OR			
	b.	Explain variables and data types in C.	L2	CO3	**05**
4.	**a.**	Explain how a 1-Dimensional array is used in C with an example.	L2	CO3	**05**
		OR			
	b.	Write a program to sort an array using bubble sort.	L2	CO3	**05**
5.	**a.**	Explain the difference between strcpy() and strcat() with examples.	L3	CO3	**05**
		OR			
	b.	Write a program to demonstrate pass by value.	L3	CO3	**05**
6.	**a.**	Explain in detail STRUCTURES with examples.	L3	CO4	**05**
		OR			
	b.	What is recursion? Give an example.	L3	CO4	**05**

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7.	**a.**	Explain the concept of Pointers in C with example.	L2	CO4	**05**
		OR			
	b.	Explain the difference between fseek() and rewind() with examples.	L2	CO4	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Describe the components of computer architecture with reference to the processor, memory, and I/O devices.	L3	CO4	**10**
		OR			
	b.	Explain the role of compilers and interpreters in programming languages.	L3	CO4	**10**
9.	**a.**	Explain the types of operators in C with examples.	L3	CO4	**10**
		OR			
	b.	Write a program in C to demonstrate the use of switch and case statement.	L3	CO4	**10**
10.	**a.**	Write a program to sort a string in alphabetical order using bubble sort.	L3	CO3	**10**
		OR			
	b.	Explain how a two dimensional array is used in C with example.	L4	CO3	**10**
11.	**a.**	Explain the concept of storage classes with examples for each type.	L4	CO3	**10**
		OR			
	b.	Write a program to declare a structure to store details of a student (name, roll, marks). Use functions to: Input the details, Display them.	L4	CO3	**10**
12.	**a.**	Explain the concept of NULL pointers and their use. Write a program to demonstrate NULL pointer handling.	L2	CO3	**10**
		OR			
	b.	Write a program to find the largest element in an array using pointers and display its address.	L2	CO4	**10**
13.	**a.**	Demonstrate the Binary Search Program	L2	CO4	**10**
		OR			
	b.	Explain in brief the difference between **while** and **do–while** statement with example.	L2	CO4	**10**